$$f(tx,ty)=\frac{tx+ty}{\sqrt{tx}+\sqrt{ty}}=t^{1/2}f(x,y),\ \forall\ x,y,t\geq 0\,.$$

Thus f is homogeneous with degree $\frac{1}{2}$, and so by Euler's Theorem we have

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=\frac{1}{2}f(x,y)\,.$$

Now substituting $f=\sin u$ in the above equation, we obtain

$$x\frac{\partial(\sin u)}{\partial x}+y\frac{\partial(\sin u)}{\partial y}=\frac{1}{2}\sin u$$

$$x\cos u\frac{\partial u}{\partial x}+y\cos u\frac{\partial u}{\partial y}=\frac{1}{2}\sin u \qquad \text{... (19)}$$

Dividing both sides by $\cos u$ we obtain

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u\,.$$

Note:

Solving this problem by direct calculation will be possible; but will involve lengthy calculations.

EXERCISE 8.7

1. In each of the following cases, determine whether the following function is homogeneous or not. If it is so, find the degree.

   (i) $f(x,y)=x^2y+6x^3+7$ (ii) $h(x,y)=\dfrac{6x^2y^3-\pi y^5+9x^4y}{2020x^2+2019y^2}$

   (iii) $g(x,y,z)=\dfrac{\sqrt{3x^2+5y^2+z^2}}{4x+7y}$ (iv) $U(x,y,z)=xy+\sin\left(\dfrac{y^2-2z^2}{xy}\right)$.

2. Prove that $f(x,y)=x^3-2x^2y+3xy^2+y^3$ is homogeneous; what is the degree? Verify Euler's Theorem for f.

3. Prove that $g(x,y)=x\log\left(\dfrac{y}{x}\right)$ is homogeneous; what is the degree? Verify Euler's Theorem for g.

4. If $u(x,y)=\dfrac{x^2+y^2}{\sqrt{x+y}}$, prove that $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=\dfrac{3}{2}u$.

5. If $v(x,y)=\log\left(\dfrac{x^2+y^2}{x+y}\right)$, prove that $x\dfrac{\partial v}{\partial x}+y\dfrac{\partial v}{\partial y}=1$.

6. If $w(x,y,z)=\log\left(\dfrac{5x^3y^4+7y^2xz^4-75y^3z^4}{x^2+y^2}\right)$, find $x\dfrac{\partial w}{\partial x}+y\dfrac{\partial w}{\partial y}+z\dfrac{\partial w}{\partial z}$.

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